

Beyond the Sigmoid Function

So far, we have only used the **sigmoid activation function** ($g(z) = \frac{1}{1+e^{-z}}$). This was a natural starting point because it's the foundation of logistic regression.

However, the sigmoid function is not always the best choice. Its output is always between 0 and 1, which is useful for probabilities (like in an output layer) but can be too restrictive for hidden layers.

- **Motivation:** Consider a hidden neuron trying to learn the "awareness" of a T-shirt. Awareness isn't necessarily a binary (0/1) value or a probability. It could be a number that grows (e.g., from 0 to a very large value representing "gone viral"). A sigmoid function cannot represent this.
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The ReLU Activation Function

To model values that can be zero or any positive number, a different activation function is more effective: **ReLU (Rectified Linear Unit)**.

- **Definition:** The ReLU function is defined as $g(z) = \max(0, z)$.
 - **How it Works:**
 - If the input z is negative, the output is 0.
 - If the input z is positive, the output is just z .
 - **The Graph:** It's a two-part function: a flat line at 0 for all negative inputs, and a 45-degree line for all positive inputs.
 - **Output Range:** The output of a ReLU neuron is always 0 or a positive number (it is **non-negative**). This makes it an excellent default choice for hidden layers.
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Commonly Used Activation Functions

Here are the three most common activation functions you will encounter:

1. Sigmoid:

- **Formula:** $g(z) = \frac{1}{1+e^{-z}}$
- **Output Range:** 0 to 1.
- **Use Case:** Good for the **output layer** in a **binary classification** problem (where you need a probability).

2. ReLU (Rectified Linear Unit):

- **Formula:** $g(z) = \max(0, z)$
- **Output Range:** 0 to ∞ (non-negative).

- **Use Case:** The most common and effective default choice for **hidden layers**.

3. Linear:

- **Formula:** $g(z) = z$
- **Output Range:** $-\infty$ to $+\infty$ (any number).
- **Note:** This is often referred to as "no activation function" since the output a is simply equal to the input z .
- **Use Case:** Good for the **output layer** in a **regression** problem (where you need to predict a number, not a probability).

*(A fourth function, **softmax**, will be introduced later for multi-class classification.)*